Jonathan Sarmiento

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Education

University of Washington – Seattle, Washington **September 2021 – June 2025**
Bachelor of Science in Mechanical Engineering GPA 3.9/4.0

Technical Skills

Software
- SolidWorks
- CATIA V5
- MSC NASTRAN/Patran
- Altair Hyper Works

Frameworks
- Microsoft Suite
- Google Suite
- MicroPython
- LaTeX

Coding
- Python
- MATLAB
- HTML/CSS/JavaScript
- Git/GitHub

Other Skills
- Spanish (Fluent)
- 3D Printing
- Data Visualization
- Problem Solving

Work Experience

Boeing **Everett, WA**
Stowbins & Secondary Supports Stress Engineering Intern June 2024 – September 2024
- Executed FEM analysis of a platform using HyperMesh, Nastran, and Patran to assess failure modes; analysis proved a safe maximum allowable mechanic weight of 200 lbs, enhancing access to hard-to-reach equipment and projecting an increase in maintenance efficiency of up to 15%.
- Evaluated thousands of interface loads from FEM postprocessing for FAA compliance, assessing 14+ load cases per location; leveraged Excel to validate loads and locations against baseline values, recommending approval or rejection and identifying potential design issues.

Boeing **Renton, WA**
Cargo Systems Design Engineering Intern June 2023 – September 2023
- Redesigned 737-10 cargo clearance markers in CATIA, validating form, fit, and function; integrated changes into PLM via ENOVIA, meeting customer specifications and ensuring smooth product lifecycle management.
- Corrected engineering drawing errors in CATIA, updated PDM, and secured approvals, improving documentation accuracy and preventing delays.

Boeing **Everett, WA**
Seat Design Engineering Intern June 2022 – September 2022
- Collaborated with 5 interns to extract and universalize seat technical content from 777/787/737 documents into a standalone Seat Technical Specification (STS), enhancing consistency and accessibility of information by over 30%.
- Contributed to drafting revisions of the passenger seat design and installation criteria, pertaining to seat jack interface provisions.

HatchMed **Seattle, WA**
Engineering Intern January 2022 – April 2022
- Responsible for the production, functionality, and quality inspection of the patented medical technology referred to as “Hall Monitor”, an interactive device that displays critical information about a patient.
- Exceeded manufacturing expectations by over 70% through efficient task delegation and improved production processes. Notably developed a custom-made jig to improve alignment accuracy and reduce assembly time.

Projects

Robotic Manipulator
- Collaborated with 3 peers to build a robotic manipulator with 4 degrees of freedom, integrating CAD-modeled 3D-printed parts, servo motors, and ESP32.
- Led 3D printing of custom components, developed Python scripts for servo control, and engineered ESP32 breadboard circuitry with servo connections

Champions League Data Analysis
- Executed data analysis on UEFA Champions League datasets using Python (Pandas, NumPy), uncovering trends and crafting visualizations with Matplotlib and Seaborn.
- Streamlined data preprocessing and applied statistical methods to extract actionable insights.

Relevant Coursework

- Thermodynamics
- Mechanics of Materials
- Heat Transfer
- Machine Design Analysis
- System Dynamics
- Kinematics & Dynamics
- Fluid Mechanics
- Visualization & CAD

Leadership

Husky Run Club **Seattle, WA**
Technology Officer February 2024 – Present
- Managed website operations to ensure seamless functionality and implemented updates; utilized Git/GitHub for version control and leveraged self-taught skills in HTML, CSS, and JavaScript to fulfill duties.
- Assisted in facilitating and leading group runs, guiding participants on safe routes and cultivating a positive welcoming atmosphere.